

■ Dot

$a.b.c$ or `Dot[a, b, c]` gives products of vectors, matrices and tensors.

$a.b$ gives an explicit result when a and b are lists with appropriate dimensions. It contracts the last direction in a with the first direction in b . ■ Various applications of `Dot`:

$\{a_1, a_2\} \cdot \{b_1, b_2\}$	scalar product of vectors
$\{a_1, a_2\} \cdot \{\{m_{11}, m_{12}\}, \{m_{21}, m_{22}\}\}$	product of a vector with a matrix
$\{\{m_{11}, m_{12}\}, \{m_{21}, m_{22}\}\} \cdot \{a_1, a_2\}$	product of a matrix with a vector
$\{\{m_{11}, m_{12}\}, \{m_{21}, m_{22}\}\} \cdot \{\{n_{11}, n_{12}\}, \{n_{21}, n_{22}\}\}$	product of two matrices

■ Example: $\{a, b\} \cdot \{c, d\} \longrightarrow a c + b d$. ■ $\{\{a, b\}, \{c, d\}\} \cdot \{x, y\} \longrightarrow \{a x + b y, c x + d y\}$.

■ The result of applying `Dot` to two tensors $T_{i_1 i_2 \dots i_n}$ and $U_{j_1 j_2 \dots j_m}$ is the tensor $\sum_k T_{i_1 i_2 \dots i_{n-1} k} U_{k j_2 \dots j_m}$. `Dot` effectively contracts the last index of the first tensor with the first index of the second tensor. Applying `Dot` to a rank n tensor and a rank m tensor gives a rank $n + m - 1$ tensor. ■ When its arguments are not lists, `Dot` remains unevaluated. It has the attribute `Flat`. ■ See pages 94 and 442. ■ See also: `Inner`, `Outer`, `NonCommutativeMultiply`.